

Milk Distribution SaaS - Enterprise Product Bible v5

Comprehensive Product, Business & Technical Blueprint

Executive Summary

Multi-tenant SaaS platform connecting distributors and customers for milk and dairy subscriptions.

Business Model

Platform subscription fees for distributors, optional transaction-based revenue, premium modules and future marketplace commissions.

User Roles

Super Admin, Distributor, Customer.

Admin PRD

Verification workflows, SaaS plans, monitoring, billing oversight, distributor lifecycle management, platform settings.

Distributor PRD

Pricing, fat-based products, delivery slots, customer onboarding, subscriptions, billing, collections, notifications.

Customer PRD

Distributor discovery, subscriptions, pause requests, extra milk requests, billing and communication.

RBAC Matrix

Detailed role-based access control for every module and action.

Subscription Engine

Daily, alternate-day, weekdays, weekly, monthly plans; pause rules; cutoff windows; future custom schedules.

Delivery Engine

Generate delivery lists, route planning foundation, delivery status tracking.

Billing Engine

Usage-based billing from deliveries, payment tracking, dues, billing cycles.

Notification Engine

In-app, SMS, WhatsApp, email-ready architecture.

Database Blueprint

Users, distributor profiles, customer profiles, pricing, products, slots, subscriptions, deliveries, bills, payments, notifications.

API Domains

Auth, Distributor, Customer, Subscription, Billing, Delivery, Notification, Admin.

Security

JWT, refresh tokens, RBAC, audit trails, validation, approval workflows.

Testing Strategy

Unit tests, integration tests, E2E tests, UAT, load testing.

Deployment Strategy

Frontend, backend, database, caching, monitoring and backup considerations.

KPIs

Revenue, active subscriptions, churn, delivery success rate, distributor growth.

Future Roadmap

Payments, wallets, mobile apps, IoT, route optimization, analytics, loyalty systems.

Module-Wise MVP to Enterprise Roadmap

Phase 1: Authentication, Distributor Setup, Customer Discovery, Subscription Management.

Phase 2: Delivery Operations, Billing Automation, Notifications.

Phase 3: Payments, Wallets, Mobile Applications, OTP Login.

Phase 4: Route Optimization, Analytics, Loyalty Programs.

Phase 5: IoT Dairy Ecosystem, Smart Devices, Enterprise Integrations.